Nonnegative and positive factorizations

Infinitesimal rigidity detects unique size-two factors in the presence of zeros

Difficulty: challenging **Importance:** interesting to specialist

Status: Lean verified

Rating rationale: Challenging because zeros change the feasible infinitesimal motions and defeat the positive-entry argument; specialist importance is a precise uniqueness test for size-two PSD factors.

Lean proof and verification evidence — 18 October 2026

The complete Lean proof establishes the displayed equivalence for every stated factorization, including all zero patterns, zero factors, singular factors and repeated rays. It retains the exact first-order trace constraint, one common positive interval of straight PSD feasibility, and every alternative factorization up to one shared genuine congruence. The actual exact-source Linux run passed Comparator, the default kernel, LeanCert kernel mode and rejection/isolation controls. Independent replay passed all 41 controls and all 90 original package inputs, including 46 Lean sources. Only the three standard axioms occur.

The reviewed statement, proof and retained evidence preserve the original target and source credit. The positive case has a direct formal quartic/conic proof. The resolution and dated audits below remain verbatim historical records; their earlier description of informal verification predates this formal proof. Exact original documents are retained.

Resolution — 2026-09-11

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For every real size-two positive semidefinite factorization of an ordinary-rank-three matrix, feasible straight-line infinitesimal rigidity is equivalent to uniqueness up to congruence. The zero-entry argument handles repeated or singular factors and zero rows and columns; the cited positive-entry theorem covers the remaining case. The feasible directions and equivalence group agree exactly with the canonical definitions.

The complete target is resolved. Its former difficulty rating is historical; the original statement, references and dated audits remain below.

Primary reference: complete authored PDF, standalone TeX, **Theorem 1, with Theorem 8 for the zero-entry construction**. Independent proof review · Authorship and submission record. Verification is independent agent review, not external human peer review or formal certification.

Context and notation

Write $\mathbb{S}_+^k$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}_+^{p\times q}$, its **real positive semidefinite rank** is

$$\operatorname{rank}_{\mathrm{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \operatorname{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of M.

Problem statement

Let $M \in \mathbb{R}_+^{p\times q}$ satisfy $\operatorname{rank}(M) = 3$ and $\operatorname{rank}_{\mathrm{psd}}(M) = 2$, and fix any size-two factorization $M_{ij} = \operatorname{tr}(A_i B_j)$. Zero entries of M are allowed.

Call a collection of real symmetric matrices (E_i, F_j) a feasible linear direction if

$$\operatorname{tr}(E_i B_j) + \operatorname{tr}(A_i F_j) = 0 \quad \text{for every } i, j$$

and some $h > 0$ satisfies $A_i + tE_i \succeq 0$, $B_j + tF_j \succeq 0$ for all i, j and $t \in [0, h)$. The trace constraint is imposed only to first order; the perturbed factors need not represent M at positive t.

Question

Are the following conditions equivalent for every such factorization?

1. Every feasible linear direction has the form $E_i = dA_i$, $F_j = -dB_j$ for one real scalar d common to all factors.
2. Every other size-two factorization $(\widetilde{A}_i, \widetilde{B}_j)$ of M satisfies $\widetilde{A}_i = S^{\mathsf{T}} A_i S$ and $\widetilde{B}_j = S^{-1} B_j S^{-\mathsf{T}}$ for one $S \in GL(2, \mathbb{R})$.

This asks whether an infinitesimal test certifies uniqueness of the optimal matrix factors up to changes of basis. Condition 1 is exactly the source's 2-infinitesimal rigidity: for size two, the relevant second-degree Taylor polynomials are the complete principal minors, and Theorem 2.2 identifies the trivial directions as the common scalings above.

Reference and status

K. Dawson, S. Hoşten, K. Kubjas, and L. Metsälampi, *Uniqueness of size-2 positive semidefinite matrix factorizations*, v2, 31 August 2026, Definition 1, Remark 1, Theorem 2.2, and the conjecture immediately after Theorem 5.2. Theorem 5.2 proves the assertion when every entry of M is strictly positive. Lemma 19 identifies local and global rigidity in the stated rank class, so omitting the equivalent local condition does not weaken the source's remaining conjecture. Searches for the title, PSD rigidity with zeros, and later work found no resolution beyond this latest version.

Status check — 2026-09-10

Rechecked Dawson–Hoşten–Kubjas–Metsälampi v2, Theorem 2.2, Theorem 5.2 and its following conjecture, and searched for later rigidity results with zero entries. The August 31, 2026 revision proves equivalence for strictly positive M and conjectures its extension to matrices with zeros. Those positive matrices are a proved part of the displayed target. No resolution of the zero-entry remainder was located.